

HoloKit: Demonstrating an Open-Source Smartphone-Based Mixed Reality Headset for Mixed Reality Design Education

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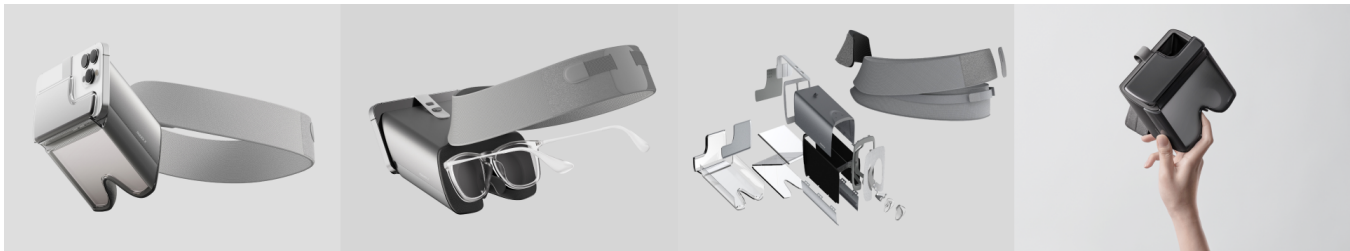


Figure 1: HoloKit: An Open-Source Smartphone-Based Mixed Reality Headset

ABSTRACT

As Mixed Reality (MR) becomes increasingly prevalent, the rising interest of students in studying MR design has led to a demand for Mixed Reality Design Education (MRDE). However, current see-through MR platforms like Vision Pro and HoloLens face educational and technological challenges: their high cost can prevent student access, and their content design, development, and authoring processes, usually tailored for professional and commercial use, can require advanced skills and be time-consuming. This discourages students from experimenting with MR designs, thus limiting their creativity and motivation to showcase their projects. To address these challenges, we introduce HoloKit, an open-source MR headset equipped with a suite of authoring toolkits. HoloKit transforms a regular smartphone into an affordable yet high-quality immersive MR experience. Targeted at MR design educators, students, and hobbyists, HoloKit is positioned as the MR equivalent of Arduino. It echoes Arduino's approach to hardware hacking and the DIY culture, offering a cost-effective alternative to more expensive MR systems. Its design philosophy emphasizes affordability, agility, and extensibility, encouraging an open-source community of creators. In this demonstration, we showcase three case studies from integrating HoloKit into real MR design classrooms and research projects. This highlights HoloKit's unique approach and effectiveness for MRDE.

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CCS CONCEPTS

• **Computing methodologies** → *Mixed / augmented reality*; • **Human-centered computing** → **Mixed / augmented reality**; *Systems and tools for interaction design*; **Ubiquitous and mobile computing systems and tools**; • **Social and professional topics** → **Computing education programs**.

KEYWORDS

Open-Source Mixed Reality Headset, Mixed Reality Design Education, Spatial Computing Education

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1 INTRODUCTION

The growth of see-through Mixed Reality (MR) platforms, such as HoloLens [23], Magic Leap [7], Meta Quest Pro [14], and the recently launched Apple Vision Pro [2], has led to increased interest in MR over the past decade. Consequently, an increasing number of students are attracted to studying MR design, resulting in higher demand for Mixed Reality Design Education (MRDE), which aims to teach students to design and develop MR experiences. MRDE is distinct from MR-Ed [31], which primarily focuses on integrating MR technologies within formal educational settings to enhance learning through interactive and immersive experiences [5, 20, 24, 25, 29]. Various subjects in MR-Ed have been researched, such as geometry [15], physics [27], chemistry [22], astronomy [6], anatomy [28], healthcare [9], art [4], design [21], architecture [8], and language [18]. However, literature specifically focused on MRDE, which involves teaching students to design and

develop MR experiences, is very limited. Despite some research on suggestions for MR design [3], teaching MR design remains a fast-growing yet immature field compared to UX/UI design education.

The latest MR devices currently face educational and technological challenges in advancing MRDE. Current see-through MR platforms, like Vision Pro and HoloLens, have high costs that can prevent student access. Their content design, development, and authoring processes, usually tailored for professional and commercial use, can require advanced skills and be time-consuming. This can discourage students from experimenting with potential MR designs, thereby limiting their creativity and motivation to showcase their projects.

In response to these challenges, we introduce HoloKit, an open-source MR headset equipped with a suite of software authoring toolkits. This transforms a regular smartphone into an affordable yet high-quality immersive MR experience. Aimed at MR design educators, students, and hobbyists, HoloKit is positioned as the MR equivalent of Arduino [13]. It echoes Arduino’s approach to hardware hacking and the DIY culture, offering a cost-effective alternative to more expensive MR systems. Its design philosophy focuses on affordability, simplicity, agility, and extensibility, which encourages an open-source community of creators to further democratize MRDE.

We conducted an experiment over two semesters involving two classes, each with 30 students. In these classes, we integrated HoloKit into their MR design curriculum. A comprehensive survey indicates that HoloKit effectively tackles challenges in the MRDE Environment to varying extents. Case studies of student and teacher projects using HoloKit confirm its success: HoloKit not only fulfills its design purpose of providing a foundational capacity for teaching MR design, but it also empowers students to create innovative projects that are unachievable on other MR platforms. Spontaneity is the most significant characteristic that differentiates HoloKit MR projects from those on other MR platforms. Examples of such projects include spontaneous collocated multiplayer MR play, contact improvisation dance MR performances, situated collocated multiplayer exhibitions, and pop-up art exhibitions engaging hundreds of audience members.

2 DESIGNING HOLOKIT HEADSET

The key idea is to utilize the existing power of the iPhone as much as possible to maximize functionality and immersiveness while keeping it affordable. Instead of reinventing the wheel, we aim to leverage the existing ARKit capabilities of smartphones. This transforms them into an affordable, yet high-quality, immersive MR experience. As illustrated in Figure 2, ARKit offers the most mature functionality among other MR ecosystems. It’s not only powerful when equipped with an iPhone that has a LiDAR scanner, but it also has abundant open-source resources and a community of developers, which can make it easier for students to create their own MR experiences.

An MR system comprises three basic components: 1) A machine perception system, which involves understanding your environment, particularly your device’s 6DOF pose, in real-time with low

Feature	ARCore	ARKit		OpenXR	
	Android	iOS	visionOS	HoloLens	Meta Quest
Session	Yes	Yes	Yes	Yes	Yes
Device tracking	Yes	Yes	Yes	Yes	Yes
Camera	Yes	Yes			Yes
Plane detection	Yes	Yes	Yes	Yes	Yes
Image tracking	Yes	Yes	Yes		
Object tracking		Yes			
Face tracking	Yes	Yes			
Body tracking		Yes			
Point clouds	Yes	Yes			
Raycasts	Yes	Yes		Yes	Yes
Anchors	Yes	Yes	Yes	Yes	Yes
Meshing		Yes	Yes	Yes	
Environment probes	Yes	Yes			
Occlusion	Yes	Yes			
Participants		Yes			

Figure 2: The table lists the available ARFoundation 5.1 [30] features for each Unity-supported MR platform plug-in.

latency and high frequency; 2) A display system, designed to convince your eyes that the virtual content is overlaid on reality; 3) An input system, allowing you to interact with the MR system.

For the machine perception system, we aim to leverage the existing iPhone Camera and LiDAR Scanner to maximize the functionality of ARKit. Unlike Attina [26], which can’t expose the iPhone’s camera set outward to enable ARKit’s full functionality, we want to fully expose the smartphone’s rear camera set facing outward, aligning it with the direction of the user’s eyesight, especially the LiDAR scanner, in order to maximize the AR functionality of our headset.

For the display system, we aim to leverage the existing OLED display of the iPhone. This allows for a very bright and high-resolution display. Furthermore, the OLED offers a non-emissive black color, which leads to complete transparency in MR, perfect for an optical see-through experience. It contrasts with a video see-through experience like Google Cardboard or Zapbox [32], which would introduce significant latency between you and reality, causing motion sickness for long wear.

For the input system, we aim to leverage the existing hand-tracking feature from the Apple Vision framework in iOS for hand-tracking input. The Apple Vision framework [1] offers a robust body and hand pose detection system powered by Apple Silicon’s Neural engine. This system can achieve real-time hand tracking at 30 FPS on the iPhone 15 Pro.

2.1 Periscope-like Z-structured core optics

Combining the key ideas above, the only place on your face to place your phone without obstructing your view is your forehead. Then we invented a periscope-like Z-structured core optics depicted in Figure 3. This allows us to fully expose the camera set outward while utilizing the iPhone’s display to combine the virtual image with reality in front of your eye.

This design consists of three optical components, each positioned at a precise 45-degree angle relative to the others, much like a periscope. The virtual image from the iPhone display is reflected off

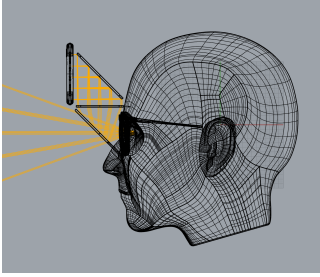


Figure 3: Periscope-like Z-structured optics

the top First Surface Reflective Mirror, passes through the center of the Fresnel lens, and merges with direct light from reality on the Beam Splitter before reaching the user's eyes.

2.2 Ergonomics and Eye Positioning

Since HoloKit positions the weight of the phone on the forehead, maintaining a balanced weight is crucial for comfortable wear. All ergonomic components, such as the headband, head pad, face mask, and nose pad, are designed not only for comfort but also to ensure the user's eye is correctly positioned within the optimal optical eye box. This design accounts for potential errors caused by the user's intense movement.

2.3 Form Factor and Aesthetics

As shown in Figure 1, our design approach for the HoloKit headset was largely influenced by the philosophy of simplicity, a characteristic beautifully reflected in the iPhone's design. Our goal was to develop a headset that not only integrates effortlessly with the iPhone but also complements its aesthetic value. We aimed for a design that would be distinctive and memorable, symbolizing HoloKit's unique periscope Z-shape. We wanted to build on the iPhone's design language, not go against it. To achieve this, we worked on creating a unified visual identity that is both straightforward and visually appealing.

3 HOLOKIT AUTHORIZING TOOLKITS

3.1 HoloKit SDK

The HoloKit SDK forms the core of the HoloKit MR platform. It turns a standard smartphone into an immersive and feature-rich MR experience. The key components include Stereoscopic Renderings, Low Latency Tracking, Hand Tracking and Gesture Recognition, and Monoscopic Video Recording during Stereoscopic renderings.

The software architecture of HoloKit SDK is built upon the existing ARFoundation ecosystem within Unity. This allows HoloKit to inherit all of ARKit's sophisticated functionalities without the need to recreate foundational technologies. This strategy ensures that HoloKit users can efficiently utilize the full spectrum of ARKit's capabilities, as shown in Figure 2.

Beyond the ARKit ecosystem, which provides features like device tracking, plane detection, image tracking, object tracking, body tracking, raycast, real-time mesh generation, and real-time occlusion, this software architecture also offers access to powerful Unity resources. These include its open-source community, Unity Visual

Effect Graph, Shader Graph, the asset store, and other open-source resources. This is an excellent starting point for MRDE students to imitate, assemble, and modify to create their own projects.

3.2 Collocated Multiplayer Boilerplate

Compared to other expensive MR devices, HoloKit's affordability makes collaborative multiplayer MR experiences accessible to MRDE students. To assist students without a strong coding or network coding background, we've developed a user-friendly collocated multiplayer MR system boilerplate [10]. This system quickly synchronizes sessions among devices, including smartphones and HoloKit, using Apple's MultipeerConnectivity, the same technology behind iPhone's AirDrop. This technology automatically connects nearby iOS devices via Bluetooth and WiFi to form a local network without requiring Internet or cellular data. This makes connections possible even in remote locations. It also offers low latency, around 30ms, which is excellent for the dynamic nature of spontaneous MR experiences. Our boilerplates use Unity's Netcode for GameObjects for network object management and MultipeerConnectivity for network transportation. Consequently, this boilerplate allows students to easily start their multiplayer MR projects without the need for complex pre-setups or pre-calibration in labs, making it very straightforward and spontaneous in the wild.

4 DEMONSTRATIONS IN MRDE CONTEXT

We conducted an experiment over two semesters involving two classes, each with 30 students by integrating HoloKit into their MR design curriculum, covering a broad spectrum of education, from idealization to realization. This process is primarily project-based, allowing students to gain hands-on experience. In these projects, students usually group together to participate in a comprehensive design project. There are various types of design projects in MRDE that students can undertake. These include MR interactive arts, MR gameplays, motion graphics design in MR, performing arts in MR, MR pop-up art exhibitions, and MR situated art exhibitions. This design journey involves numerous phases and iterations of the design process, which includes empathizing, defining, ideating, prototyping, testing, deploying, and finally exhibiting their work, as characterized by design thinking [16].

HoloKit's capabilities facilitate rapid prototyping and iterative design in educational settings and professional workshops. By lowering technical barriers and minimizing setup time, HoloKit allows users to focus on creative expression and design experimentation. This is especially beneficial in an MRDE context, where students can test hypotheses and refine concepts in real-time without the delays associated with more complex systems.

4.1 Demo 1 for Spontaneous Plays – MOFA.AR

Collaborating with MRDE students who have a strong interest in designing gameplays, *MOFA.AR* [11, 12] includes a series of MR embodied play experiences set in spontaneous environments, ranging from single-player to multiplayer scenarios (Figure 4). The designers envision players as wizards navigating urban spaces, casting spells against monsters and each other. Additionally, those without MR headsets are considered "Muggles" in the setting but are still incorporated into the play.



Figure 4: The games derived from the *MOFA.AR* framework demonstrate the possibilities of multiplayer collocated MR games: (Left) *MOFA The Duel*: the two HMD wearers compete by throwing virtual magic energy balls at each other while avoiding getting hit; (Middle) *MOFA The Dragon*: The HMD wearers can combat the dragon agent, which is controlled by the handheld AR player.; (Right) *MOFA The Ghostbuster*: An asymmetric play where the "detector" can fire a detection wave to make the "ghost" visible, while the "ghostbuster," the HMD wearer, can throw an energy ball to harm the ghost.



Figure 5: *Cell Space*, based on HoloKit Collocated Multiplayer Boilerplate; (Left) and (Middle): The spectator view of *Cell Space*, which experiments with augmented intercorporeal awareness through MR; (Right) Screenshot of *Cell Space* developed in Unity using HoloKit authoring tools.

HoloKit introduces spontaneous multiplayer gameplay "in-the-wild," a feature that lacks support from existing authoring tools [17]. Incorporating MR head-mounted players and handheld mobile AR players into one game in the wild presents significant challenges. HoloKit's collocated multiplayer boilerplate addresses this by offering a QR code scanning approach and leveraging the AirDrop technology supported by iOS.

4.2 Demo 2 for Performing Art – Cell Space

Inspired from *MOFA.AR*, collaborating with MRDE students who have an performing dance background, *Cell Space* [19] is a spontaneous, passive-interactive multiplayer MR performance art piece that visualizes dancers as dynamic cells within a space (Figure 5). The cells in *Cell Space* are generated procedurally based on real-time spatial data, with the dancers' positions forming sites in a 3D Laguerre-Voronoi diagram. "In the post-pandemic era, we are questioning the boundaries between people." The designer aims to raise awareness of intercorporeality and isolation. They create an intuitive affordance for dancers to realize and actively respond to each other's intentions through a new layer of perception enabled by bio-sensing technology and augmented visuals.

Another significant challenge is debugging in collocated MR settings, especially in performing arts, which often require multiple performers and specific physical spaces—resources that are not always readily available. HoloKit provides authoring tools that let designers pre-capture real-world scenes with embedded metadata, supporting the direct design of collocated mixed reality experiences



Figure 6: *EchoVision*, a pop-up exhibition near the natural habitat of a wild bat population; (Left) The customized bat-shaped masks with HoloKit inside were hanging upside down on the wall before the pop-up event; (Middle) Participants and passersby tried the MR experience during the pop-up exhibition; (Right) First-person view of the *EchoVision* MR experience.

based on the physical environment. By integrating AI-based human detection and depth segmentation, along with the capabilities of ARKit and LiDAR, without being compiled into an actual MR device, it can render the final composited AR videos in the studio, overcoming barriers of time-consuming multi-user debugging. Additionally, the adoption of Unity's Universal Render Pipeline (URP) ensures that the visual fidelity of MR content on HoloKit closely matches what is seen on the screen in the studio. This facilitates a seamless transition from rapid prototyping to the final realization of performing arts projects.

4.3 Demo 3 for Pop-Up Exhibition – EchoVision

Collaborating with MRDE students who have an industrial design background, *EchoVision* is a pop-up art event in-the-wild (Figure 6), offering a spontaneous MR active experience that transforms the player's vision into that of a bat, in an artistic expression. Players can trigger visual echoes with the physical environment through the control of their voice.

The design goal of *EchoVision* is to attract audiences to try it out without needing instructions on how to use it, as pop-up events often host large numbers of people in an uncontrolled environment. The customizability of HoloKit allows it to be enhanced with 3D-printed components, making it both attractive and intuitive for users to engage with. HoloKit's spontaneity caters to hundreds of people, as participants are eager to engage with *EchoVision* while watching others play and enjoy the experience.

5 CONCLUSION

These demonstrations from our actual MRDE teaching process show the potential of HoloKit as a transformative tool in MRDE. By addressing the MRDE challenges through an open-source, cost-effective platform, HoloKit empowers students and educators to engage deeply with MR as a new medium. It facilitates a creative and collaborative environment that enhances learning and fosters innovation across various disciplines. The capabilities of HoloKit allow it to serve both as a learning tool and as a bridge between theoretical knowledge and practical application, enabling users to experiment and iterate on their designs rapidly. As HoloKit continues to evolve and expand its community, it promises to further democratize MR technology, making it accessible to a broader audience and paving the way for the next generation of MR designers and creators.

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